

# Unified Asset & Operations Brain - User Manual & Project Guide

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## ET - AI Hackathon (Problem Statement 8) — Official User Guide

Welcome to the **Unified Asset & Operations Brain**, an AI-powered Industrial Knowledge Intelligence and Maintenance Root Cause Analysis (RCA) platform designed for heavy industrial facilities (Northgate Energy / Northgate Refinery).

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## 1. Quick Start: How to Run the App

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Both servers run locally on your computer:

```
# Option A: Start Backend API (Python FastAPI)
cd c:\Users\dubey\Desktop\ET_AI\backend
python -m uvicorn api.main:app --host 127.0.0.1 --port 8000 --reload

# Option B: Start Frontend Web UI (React + Vite)
cd c:\Users\dubey\Desktop\ET_AI\frontend
npm.cmd run dev -- --port 5173
```

- **Web Dashboard URL:** <http://localhost:5173>
  - **FastAPI API URL:** <http://localhost:8000> (Interactive Docs: <http://localhost:8000/docs>)
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## 2. Navigating the 5 Web UI Tabs

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### Tab 1: ■ Control Overview

- **What you see:** System health badges, document ingestion counters, Knowledge Graph triple counts, and high-risk equipment heat cards (e.g. Compressor C-102, Boiler B-201).
- **Quick Action:** Click any equipment card to jump directly to its Root Cause Analysis report.

### Tab 2: ■ Expert RAG Copilot (Chatbot)

- **What you see:** A dark-mode conversational interface powered by Retrieval-Augmented Generation (RAG) over all 10 master refinery documents.
- **Features:**
- **Auto-Scroll to Answer:** Automatically aligns the view to the top of the AI's latest response so you can read immediately.

- **Quick Showcase Questions Toolbar:** Click the `Quick Showcase Questions` button at the top right to select pre-configured questions!
- **Verified Source Citations:** Every technical answer displays clickable citation pills (e.g. `01_refinery_working_and_parts_guide.txt`). Click any pill to inspect the exact retrieved text snippet in the right drawer.
- **Domain Guardrails:** If asked out-of-scope questions (e.g. *"Who won the IPL?"*), the AI politely refuses and redirects to company topics.

### Tab 3: ■ Maintenance & RCA Agent

- **What you see:** Autonomous reliability pattern analysis for plant equipment.
- **How to use:** Select any equipment tag (C-102, PRV-88, P-401B, T-101) to view its **Chronological Operational Timeline** and an **Agentic Root Cause Analysis (RCA) Report** detailing failure modes, root causes, violated SOPs, and preventative recommendations.

### Tab 4: ■■ Plant Map & Piping Network

- **What you see:** An interactive SVG industrial blueprint map of Northgate Refinery (450-acre facility).
- **How to use:** Click on any of the 6 plant zones (**Zone A: Crude Processing**, **Zone B: Hydrocracker**, **Zone C: Boiler House**, **Zone D: Tank Farm**, **Zone E: Pipe Racks**, **Zone F: Safety CCR**) to inspect equipment coordinates, grid sectors, connected pipe lines, and emergency assembly points.

### Tab 5: ■ Knowledge Base & Graph Explorer

- **What you see:** Ingested document catalog and SQLite Knowledge Graph entity triples explorer.
- **How to use:** Upload new PDFs or text files via drag-and-drop, or click entity buttons (e.g. C-102, B-201, OISD-116) to inspect relationship triples.

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## 3. How to Demonstrate the App to Friends & Judges

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For maximum impact during a presentation:

1. **Start on the Plant Map Tab:** Show the 6 plant zones and explain that the refinery processes 15.2 MMTPA of crude oil.
2. **Open the RAG Copilot Tab:**
3. Click `Quick Showcase Questions` and ask *"can u give me a map"*.
4. Ask *"What is the required Octane rating for BS-VI petrol?"* and click the source citation pill to show the raw document snippet in the drawer!
5. Ask *"Who won the IPL?"* to demonstrate AI domain guardrails!

6. **Open the RCA Agent Tab:** Select C-102 and show how the AI extracted a 2018–2024 failure timeline and identified monsoon moisture ingress as the root cause.
7. **Share the PDF Guides:** Show the PDF guides in the PDF\_GUIDES/ folder!

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## 4. PDF Guides Index in PDF\_GUIDES/ Folder

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All PDF guides are neatly organized in c:\Users\dubey\Desktop\ET\_AI\PDF\_GUIDES\:

1. 01\_Oil\_Refinery\_Construction\_and\_Working\_Guide.pdf: Beginner-friendly guide explaining how an oil refinery works, all 7 sections, and construction phases.
2. 02\_30\_Meaningful\_Showcase\_Questions\_and\_Answers.pdf: 30 curated questions with expected answer previews across 6 categories.
3. 03\_Project\_Implementation\_Plan.pdf: Technical architecture, tech stack, and database schema documentation.
4. 04\_User\_Manual\_and\_Project\_Guide.pdf: This official User Manual & Demonstration Guide.